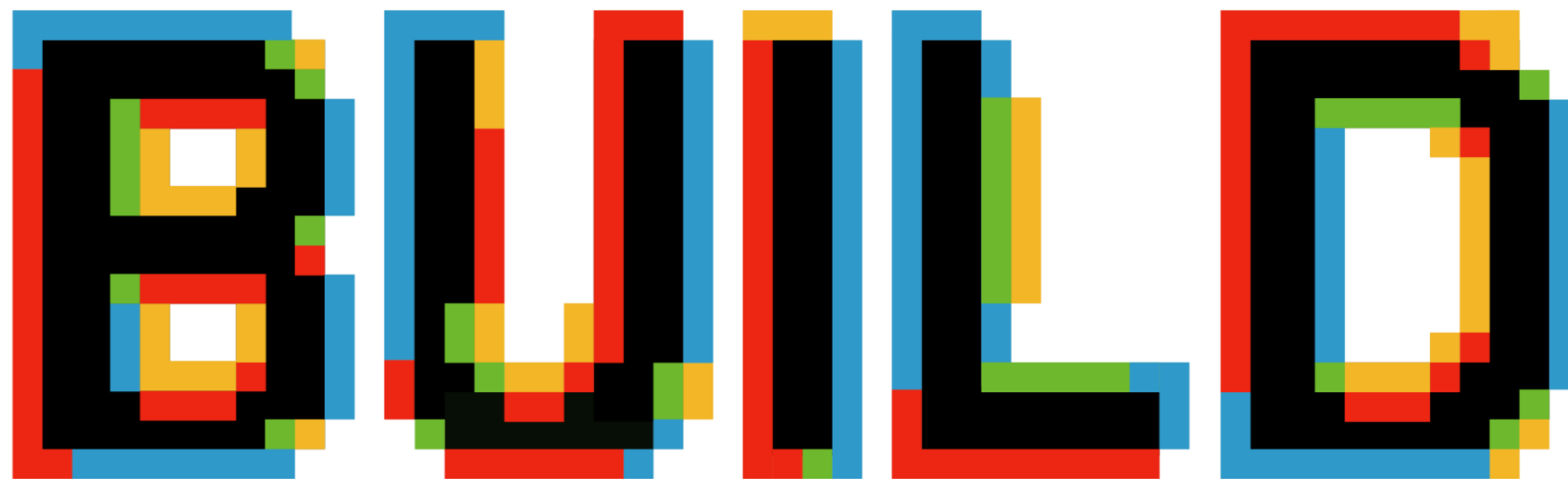
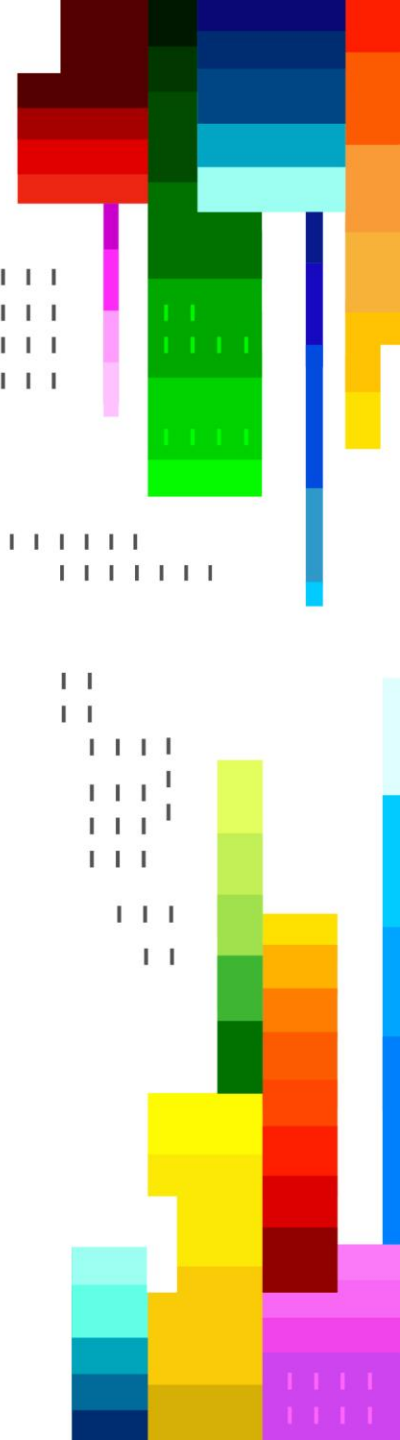


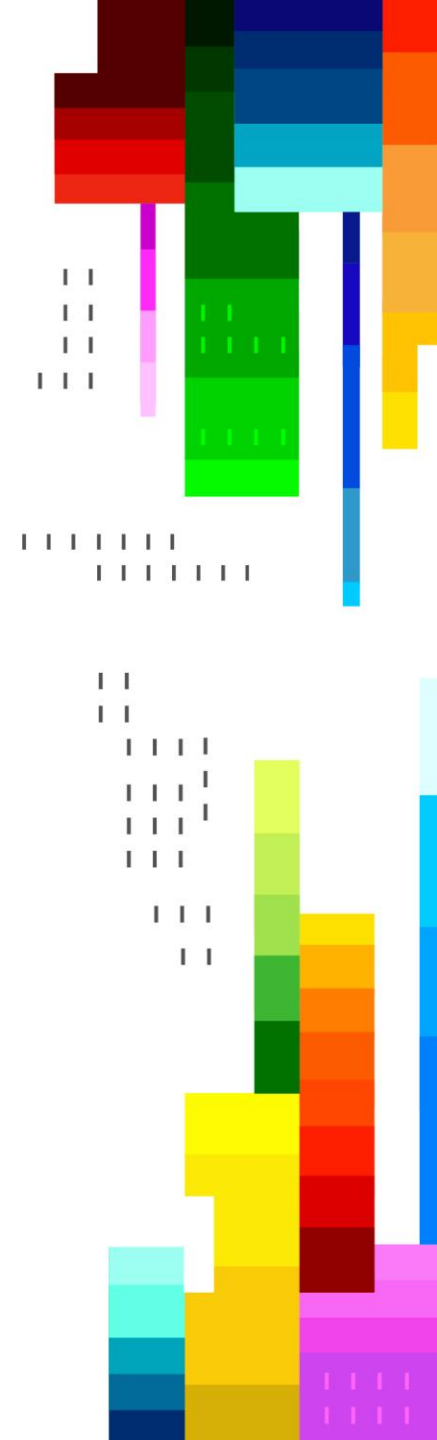


MICROSOFT BUILD



LAB511

Create advanced Postgres-powered
agentic apps with Azure HorizonDB



Meet the speakers and proctors



**Jonathon
Frost**

Principal Product
Manager, Microsoft



Denzil Ribeiro

Principal PM
Manager,
Microsoft



Danyal Bukhari

Program Manager,
Microsoft



Kanchan Bharati

Senior Program
Manager, Microsoft



Charles Feddersen

Partner Director of Product
Management, Postgres,
Microsoft



Affan Dar

VP Engineering,
Postgres, Microsoft



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THANK YOU
FOR YOUR FEEDBACK!

LAB511

Create advanced Postgres-powered
agentic apps with Azure HorizonDB

Your feedback is very useful for us and
Helps us improve our labs for future events

Lab content: Please
open an issue in the repo



<https://aka.ms/build2026-LAB511>

Agenda

-
- Lab Overview
 - HorizonDB AI Core Concepts
 - Lab "Follow Me"
 - Part 0 – Login to Azure
 - Part 1 – Connect to HorizonDB using VS Code Extension
 - Part 2 – Notebook 1: Data Setup
 - Part 3 – Notebook 2: Application Development / Build the Agentic App

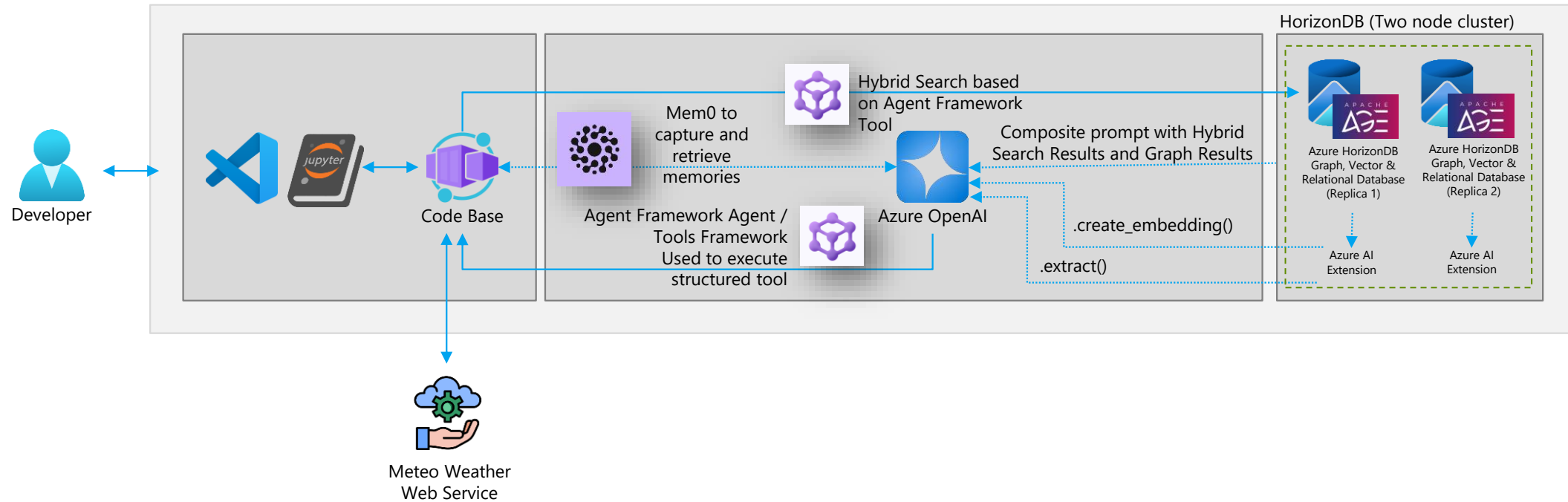
Lab Overview

What you will learn:

- How to use the VS Code PostgreSQL Extension with HorizonDB
- Understand the AI & Graph Features of HorizonDB such as
 - How to use Vectors and Vector Indexes with HorizonDB
 - How to use BM25 search with HorizonDB
 - How to use AI Functions with HorizonDB
 - How to build, query, and visualize a graph database in HorizonDB
- Understand the enterprise features of HorizonDB such as:
 - Multi-Replica Clusters
 - Fail-over and resilience coding patterns
- Learn about Agentic App architectures and coding patterns

Agentic App Architecture

The App we are going to build today



Counsel AI - Legal Research Assistant

Ask legal questions. Counsel uses HorizonDB to find, analyze, and synthesize case law.

cases, most involving property, sometimes involving harm, emphasizing many more severe consequences aggravated moving conditions.

- Cases like **Thorpe v. City of Spokane** and **Colella v. King County** confirmed that negligent grading or drainage changes resulting in property damage may support claims for damages.
- **Pruitt v. Douglas County** and **Wilber Development Corp. v. Les Rowland Constr., Inc.** recognized liability where artificial alterations exacerbated stormwater flows, causing property injury.
- **Borden v. City of Olympia** and **Currens v. Sleek** established that inverse condemnation or nuisance actions could proceed if governmental or neighbor conduct directly enabled flooding.

The cited line of authority thus recognizes claims when either a neighbor or the city changes drainage patterns, overcoming the general "common enemy" rule, if those changes are unnatural or unreasonable.

Weather Evidence: Supporting Severe Storm Impact

Between April and May 2024, your Queen Anne property weathered significant rainfall, with 97.7 mm total precipitation over 36 rainy days. This amount is consistent with a "heavy storm" and provides strong factual support for the assertion that post-regrading and repaving, your client's property suffered persistent and unnatural flooding on practically every rainy day. This evidence substantiates abnormal drainage issues—not just coincidental increased rainfall—and is critical in showing causative impact rather than just "natural conditions."

Key Takeaway:

Your client likely has a viable claim against both the neighbor (for regrading) and potentially against the city (for repaving) if you can show the flooding is due to artificial change in drainage patterns, rather than natural stormwater flow. Documentation of flooding frequency, severity, and evidence linking the grading and repaving to the onset of those issues will be central.

Key Cases in this Line of Authority:

- DiBlasi v. City of Seattle, 1997/1998 (Washington Supreme Court & Court of Appeals)
- Thorpe v. City of Spokane, 1914 (Washington Supreme Court)
- Colella v. King County, 1967 (Washington Supreme Court)
- Pruitt v. Douglas County, 2003 (Washington Court of Appeals)
- Wilber Development Corp. v. Les Rowland Constr., Inc., 1974 (Washington Supreme Court)
- Borden v. City of Olympia, 2002 (Washington Court of Appeals)
- Currens v. Sleek, 1999 (Washington Supreme Court)

Statutes Extracted:

- RCW 4.24.630 (referenced in Borden v. City of Olympia for waste and nuisance claims related to surface water diversion)

Holdings Extracted:

(Case extraction failed due to an API limitation; please consult full texts of the cases above for detailed holdings. The cases establish liability for artificial changes to drainage patterns causing harm.)

Action Steps:

- Gather evidence of pre/post flooding conditions.
- Document neighbor's grading and city's repaving (dates, methods, before/after photos).
- Consider consulting hydrology or civil engineering experts to link the drainage change to flooding.
- Assert claims under nuisance, negligence, inverse condemnation, and trespass doctrines.

The weather data and cited cases together provide a robust foundation for moving forward with your client's claim.



Tool Trace (5 tools used)

- KeywordCaseSearchTool** (BM25 Search) 416 ms
Searched for "surface water flooding AND regrading AND city repaving " and related terms.
Results: 5 cases
- SemanticCaseSearchTool** (DiskANN with Advanced Filtering) 1.6s
Expanded search using vector similarity with filters: court_level IN (Both), decision_date >= 1900.
Results: 5 cases
- PrecedentGraphTool** (AGE Graph) 2.2s
5 unique cases found via KeywordCaseSearchTool and SemanticCaseSearchTool.
Paths found via graph traversal
- CaseAnalystTool** (AI Function azure_ai_extract[]) 2.2s
Extracting structured legal entities from 10 case opinions:
holding issues statutes_cited disposition
Extracted 10 cases:
✓ Colella v. King County (id 1091252)
✓ Phillips v. King County (id 867982)
✓ Pruitt v. Douglas County (id 265811)
✓ Wilber Development Corp. v. Les Rowland Constr., Inc. (id 1127960)
✓ King County v. Boeing Co. (id 1036918)
✓ Borden v. City of Olympia (id 764681)
✓ Whiteside v. Benton County (id 701978)
✓ Tope v. King County (id 1279441)
✓ Currens v. Sleek (id 887428)
✓ Rothweiler v. Clark County (id 321060)
- WeatherEvidenceTool** (Weather External API) 10.0s
Retrieved historical precipitation data for 2024-04-01.

Completed in 18.01 seconds

Mem0 Long-Term Memory (user: ui-counsel-session2)

RETRIEVED FOR THIS TURN (2)

Client owns a property in the Queen Anne neighborhood of Seattle, Washington

Is an attorney

CAPTURED THIS TURN (4)

Client's yard floods every time it rains after a heavy storm during April and May of 2024

Neighbor regraded their lot in February 2024

City repaved the road in February 2024

Client is seeking to determine if they have a legal claim regarding the flooding

Example questions

1. Queen Anne client

2. Yard flooding after neighbor regrade & city repave

3. Most promising defendant & evidence to collect

4. Spaceship crashed into a second client's condo

Ask a legal question...

Send

App UI

Lab GitHub Repo

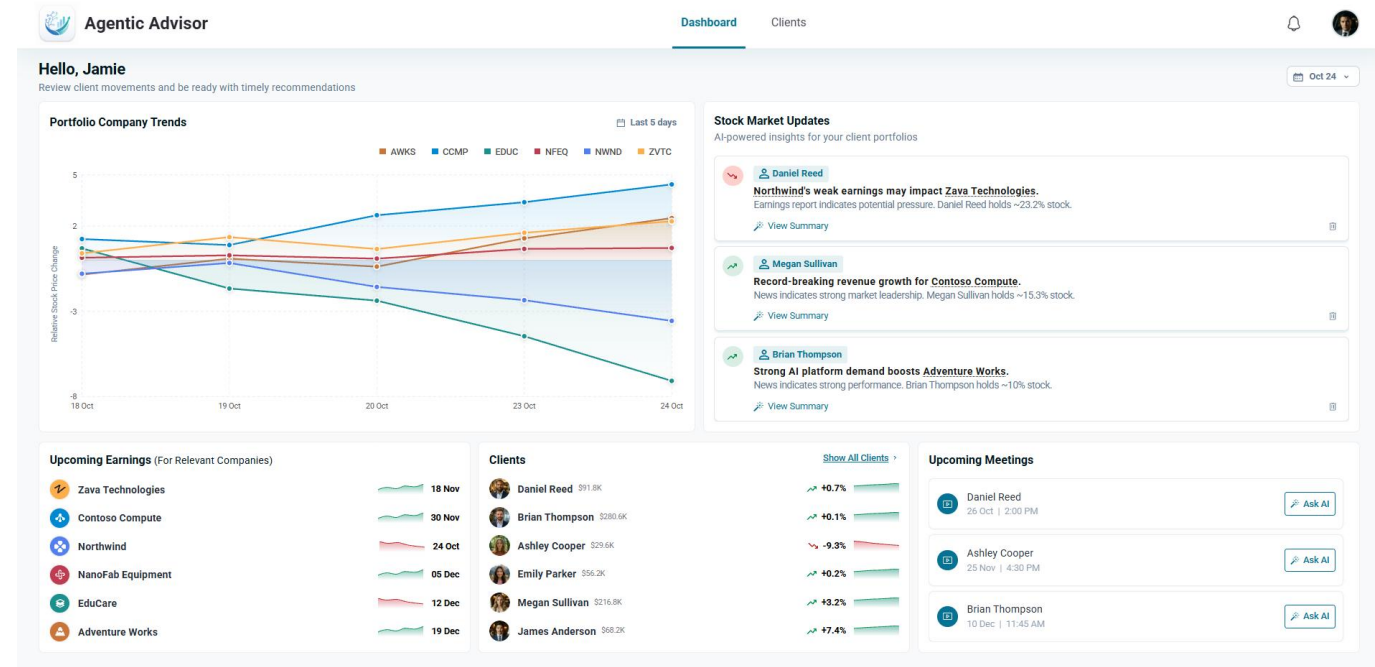
(Long term)



github.com/jjfrost/build-2026-lab

Agentic Advisor Solution Accelerator

(Uses most of the technologies from this lab)



<https://aka.ms/agentic-advisor>

Dataset for the Postgres Lab

- Caselaw Dataset for Washington State
- Subset of 337 unique legal cases
- Columns include: id, name, opinion, etc.
- Located at /Dataset/cases.csv

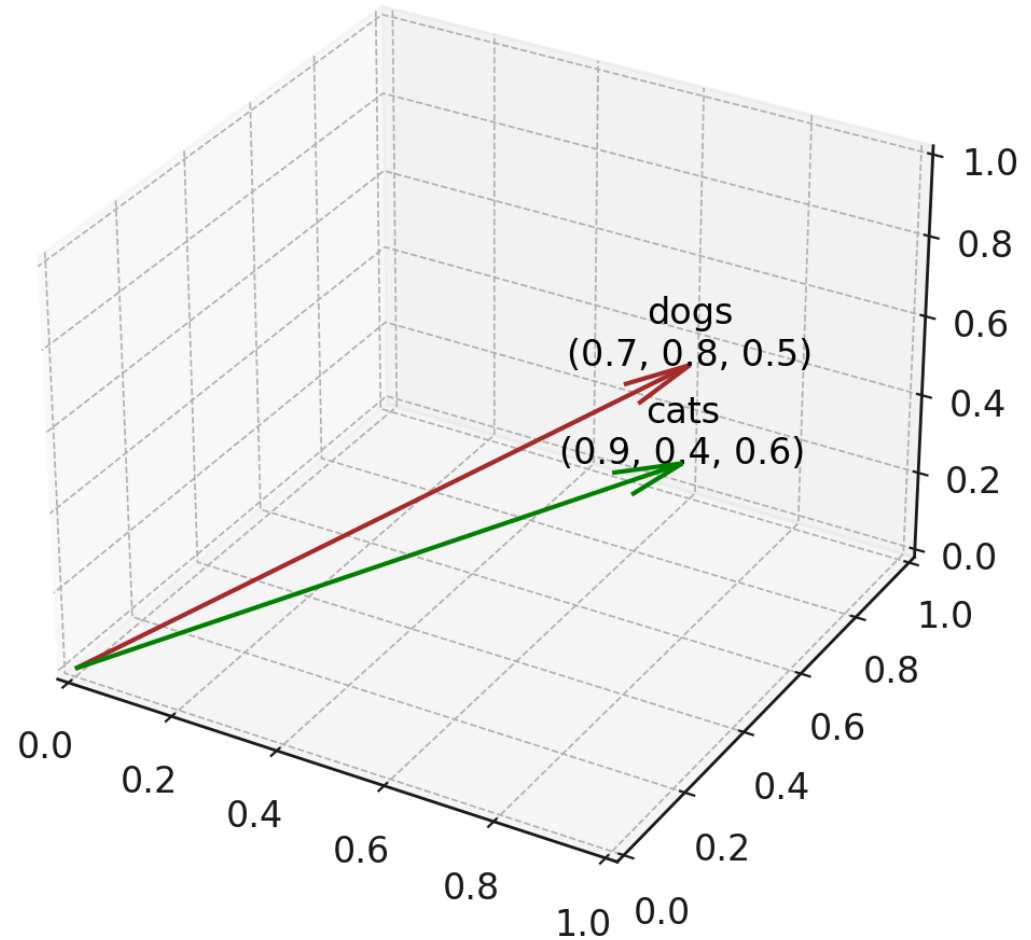
	id ↕	name ↕	decision_date ↕	court_id ↕	opinion ↕
1	507122	Berschauer/Phillips Construction Co. v. Seattle Sc...	1994-10-06	9029	"Guy, J.\nWe granted review to decide whether a gen...
2	5041745	Frisken v. Art Strand Floor Coverings, Inc.	1955-10-13	9029	"Rosellini, J.\nThe respondent, Florence Frisken, i...
3	5008733	Pate v. General Electric Co.	1953-09-04	9029	"Weaver, J.\nPlaintiff was injured while engaged in...
4	5007905	Cambro Co. v. Snook	1953-11-05	9029	"Donworth, J.\nPlaintiff instituted this action to ...
5	5008594	Buttnick v. Clothier	1953-11-16	9029	"Donworth, J.\nThis action was instituted by plaint...

AI Core Concepts

- Vector Embeddings
- Semantic Search
- Vector Indexes
- Agent Development Architecture
- GraphRAG
- HorizonDB AI Functions

What is a Vector?

- Lists of numbers that represent items in a high-dimensional space.
- For example, a vector representing the string "dogs" might be $[0.7, 0.8, 0.5]$.
- Each number in the vector is a dimension of the space.



How to generate a vector?

Use a model to generate vectors for items:

Input	→ Model	→ Vector
"dog"	text-embedding-3-small	[0.017198, -0.007493, -0.057982, ..]
"cat"	text-embedding-3-small	[0.004059, 0.06719, -0.093874, ...]

Using in our lab

Model (bi-encoder)	Input types	Dimensions
OpenAI: text-embedding-3-small	Text	1536
OpenAI: text-embedding-3-large	Text	3072
Mistral: e5-mistral-7b-instruct	Text	4096

Storing vectors in the HorizonDB 'cases' Table

Results		Messages		
	id	name	opinions_vector	
1	507122	Berschauer/Phillips Construction Co. v. Seattle Sc...	[-0.0077604363,0.034168452,0.022548927,0.058252566,0.0027358707,0.013302599,-0.04104158,-0.0011557909,-0.02792912,-0.00568652...	
2	5041745	Friskens v. Art Strand Floor Coverings, Inc.	[0.008968134,0.04363906,-0.0017264026,0.0380413,0.006953235,0.0002628528,-0.022229837,-0.028633554,-0.011818302,0.0461009,0.0...	
3	5008733	Pate v. General Electric Co.	[-0.009503542,0.052598044,-0.00058293104,0.051410984,0.013446276,0.017848289,-0.013997411,-0.02381185,-0.020533305,0.03219192...	
4	5007905	Cambro Co. v. Snook	[0.02875072,0.033727877,0.00932174,0.004737335,0.037787456,0.01634954,-0.045406118,-0.019574959,-0.010670299,0.017281018,0.03...	
5	5008594	Buttnick v. Clothier	[0.0077795624,0.035135385,0.029488107,0.02745043,-0.017844236,0.013717937,-0.023156751,-0.028396495,-0.03015763,-0.03202065,0...	

Optimize Vector Query Performance

Index Types

IVFFlat

- Clusters vectors by applying k-means clustering.
- Memory efficient but requires index rebuilds.

Generally speaking... IVFFLAT does its best work for larger datasets with low dimensionality.

HNSW

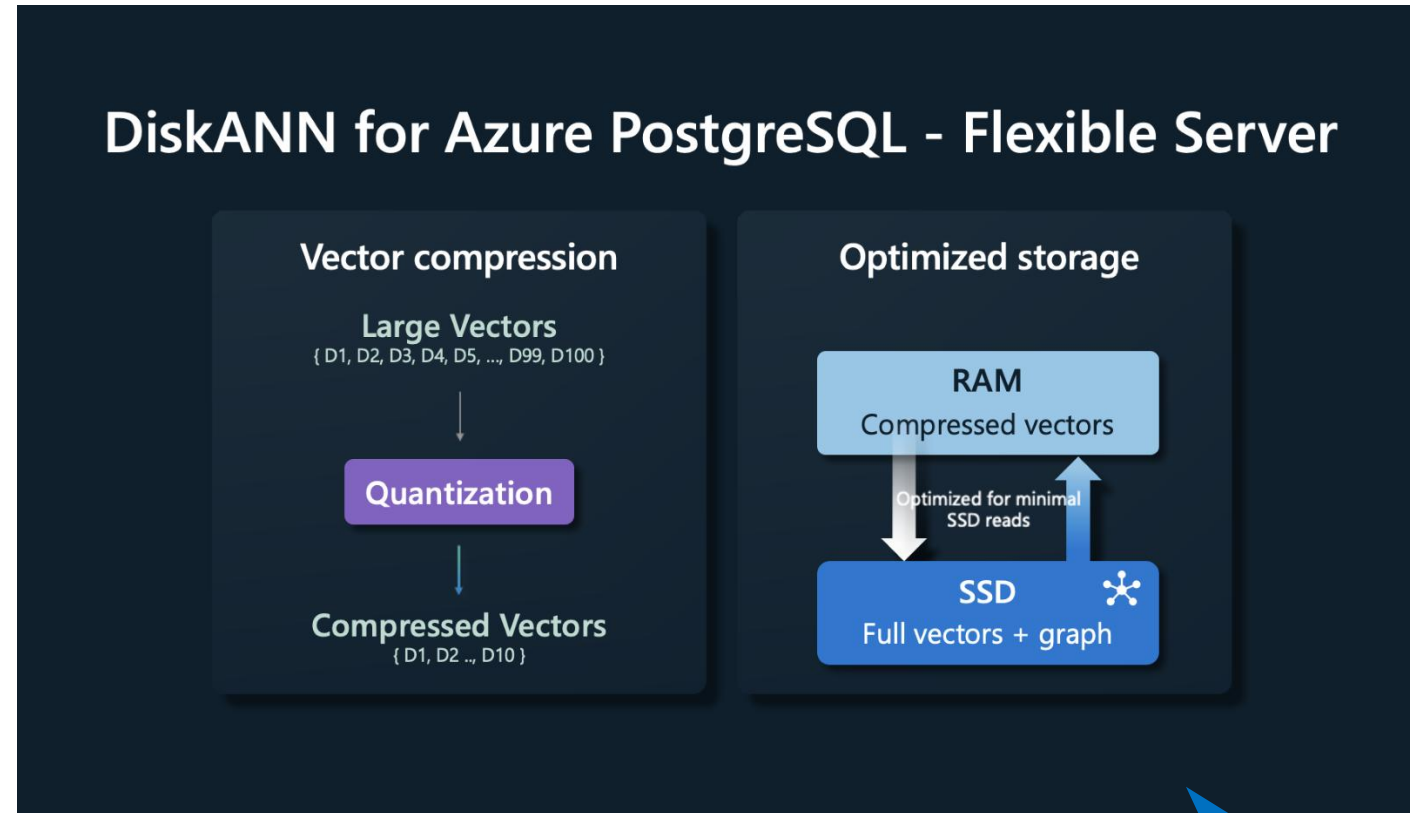
- Builds a multi-layer graph with long and short connections between the vectors.
- The graph can be incrementally updated.
- Memory hungry

Generally speaking... HNSW does its best work for smaller datasets with high dimensionality.

What do you do for large datasets with high dimensionality?

Vector Indexing - DiskANN

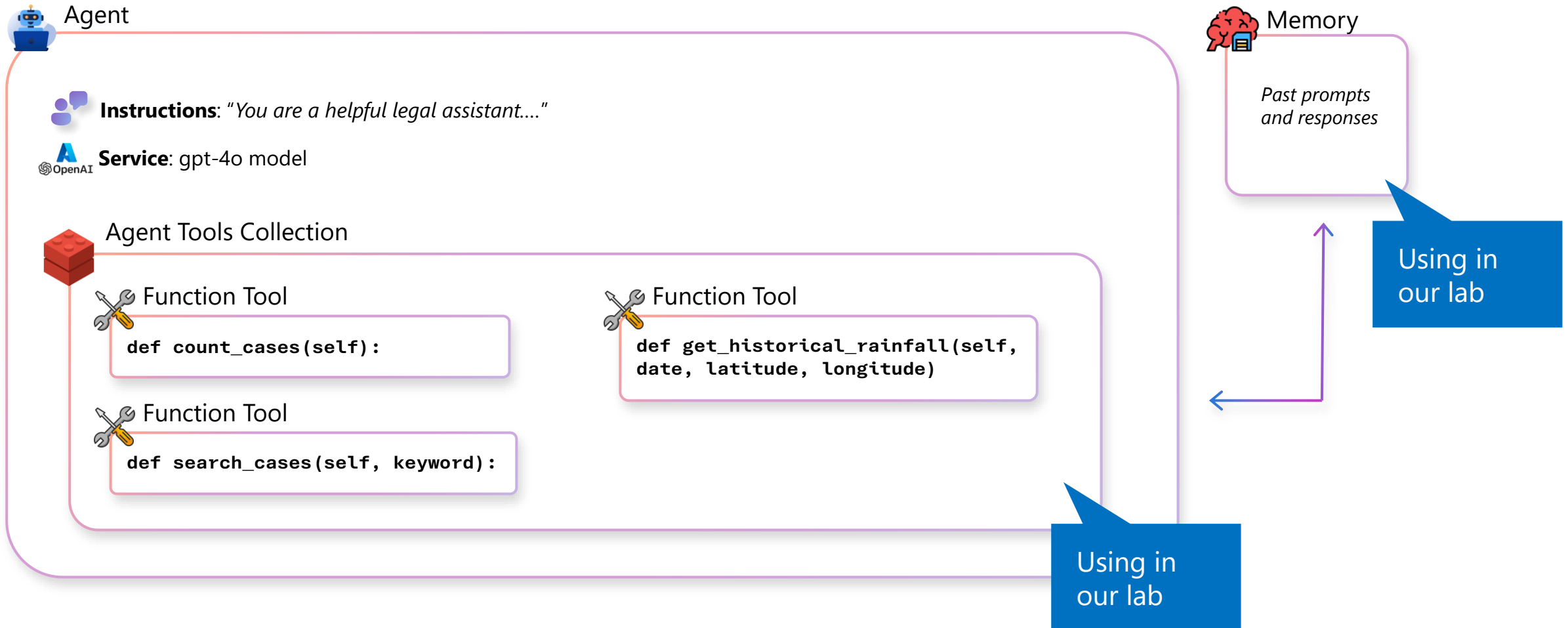
- Highly performant, scalable, and accurate index for vectors
- Superior to IVFLAT and HNSW
- Reduced memory footprint by storing vectors on SSD
- Compression and quantization improve speed and accuracy of vector search
- Accuracy retained as data changed



Using in
our lab

Agents

Agent Framework

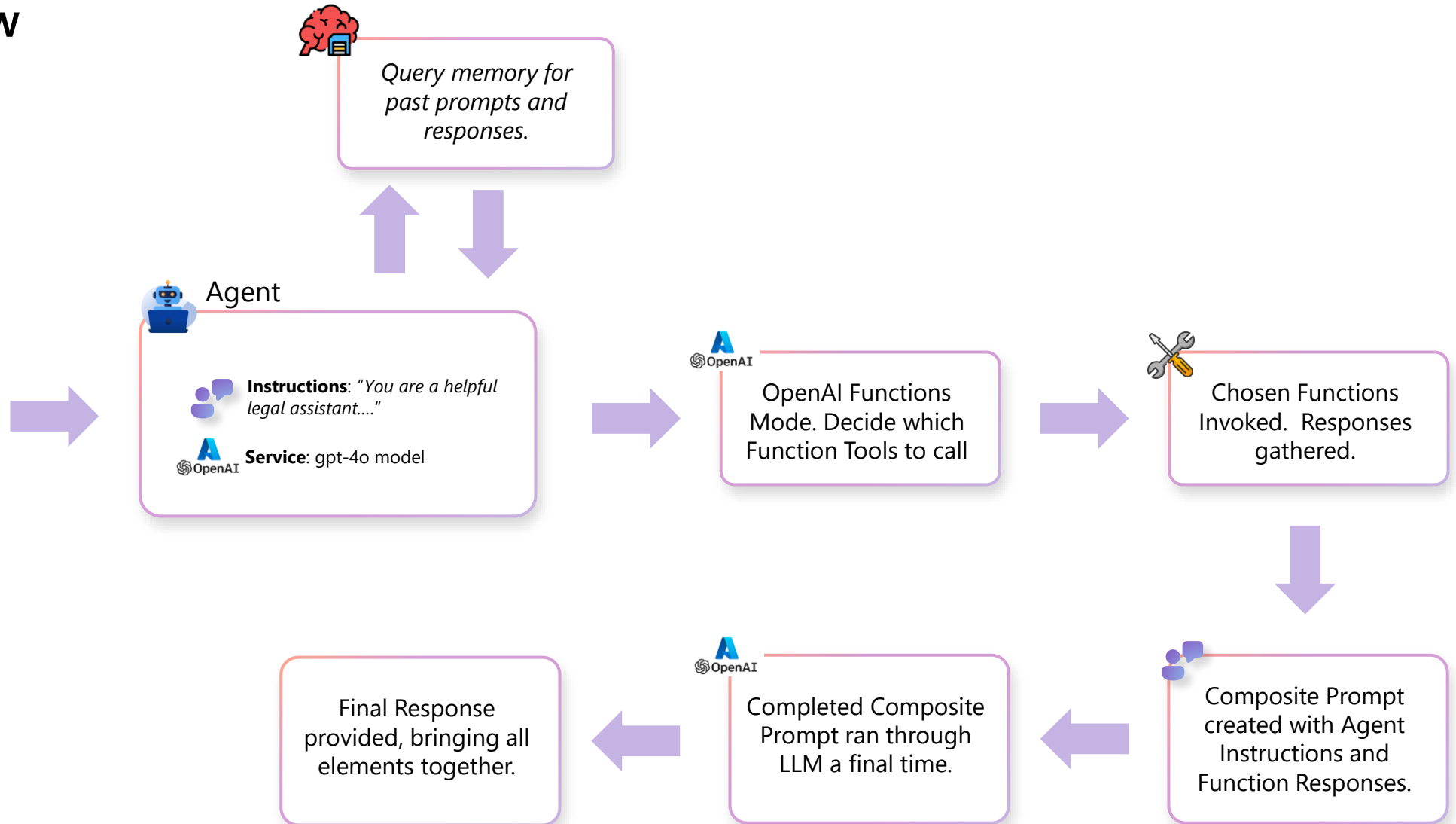


Agents

Logical Flow

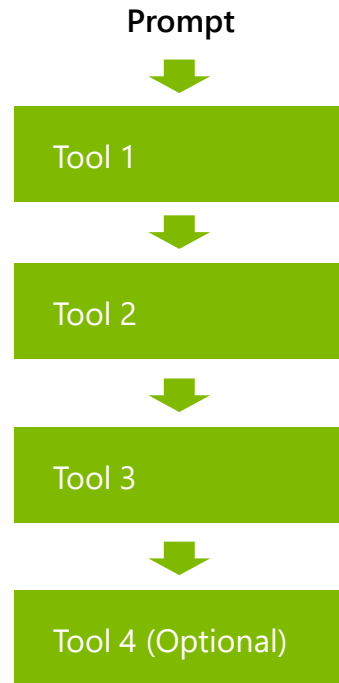
Prompt:

"How many cases are there, and high accuracy is important, help me find 5 highly relevant cases related to water leaking in client's apartment."



Agentic Application Patterns

Some examples...



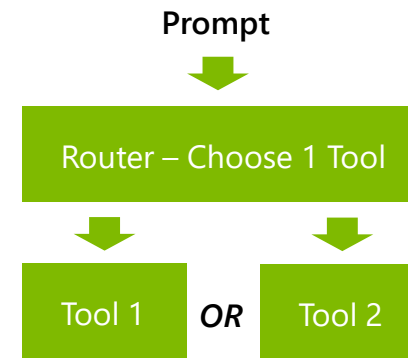
Pipeline / Chain-of-Tools

- Fixed or semi-fixed order
- Each tool transforms the output of the previous step
- Data becomes progressively more refined



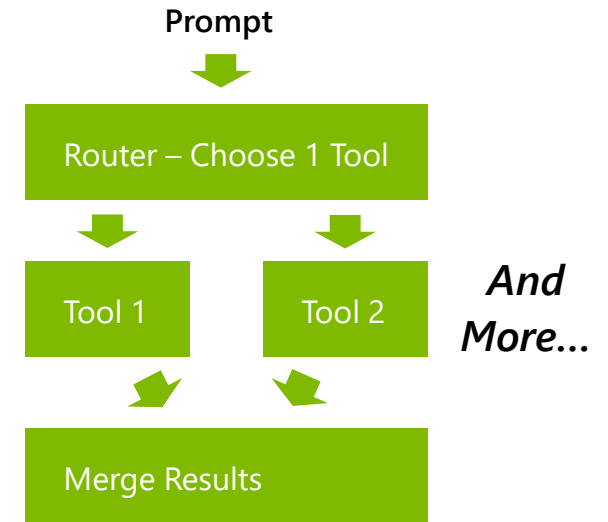
ReAct (Reason + Act Loop)

- Dynamic tool selection
- Iterative
- Non-linear



Router

- Single-step decision
- No chaining
- Good for simple assistants



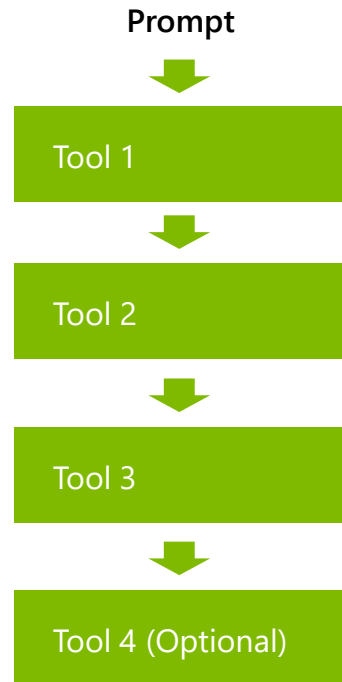
Parallel Tool Call

- Faster
- Used for independent tasks

Agentic Application Patterns

Some examples...

Using in
our lab



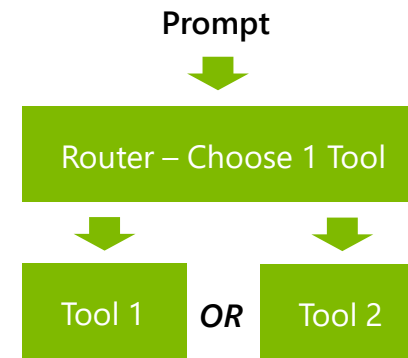
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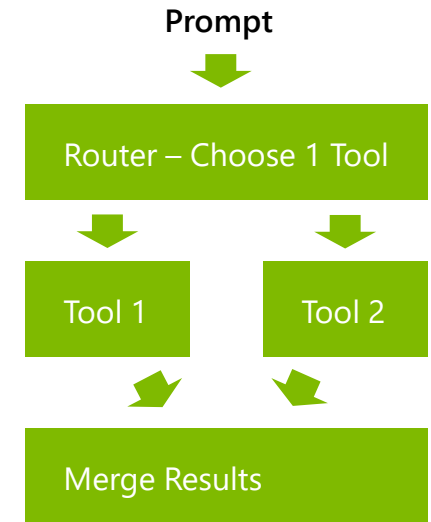
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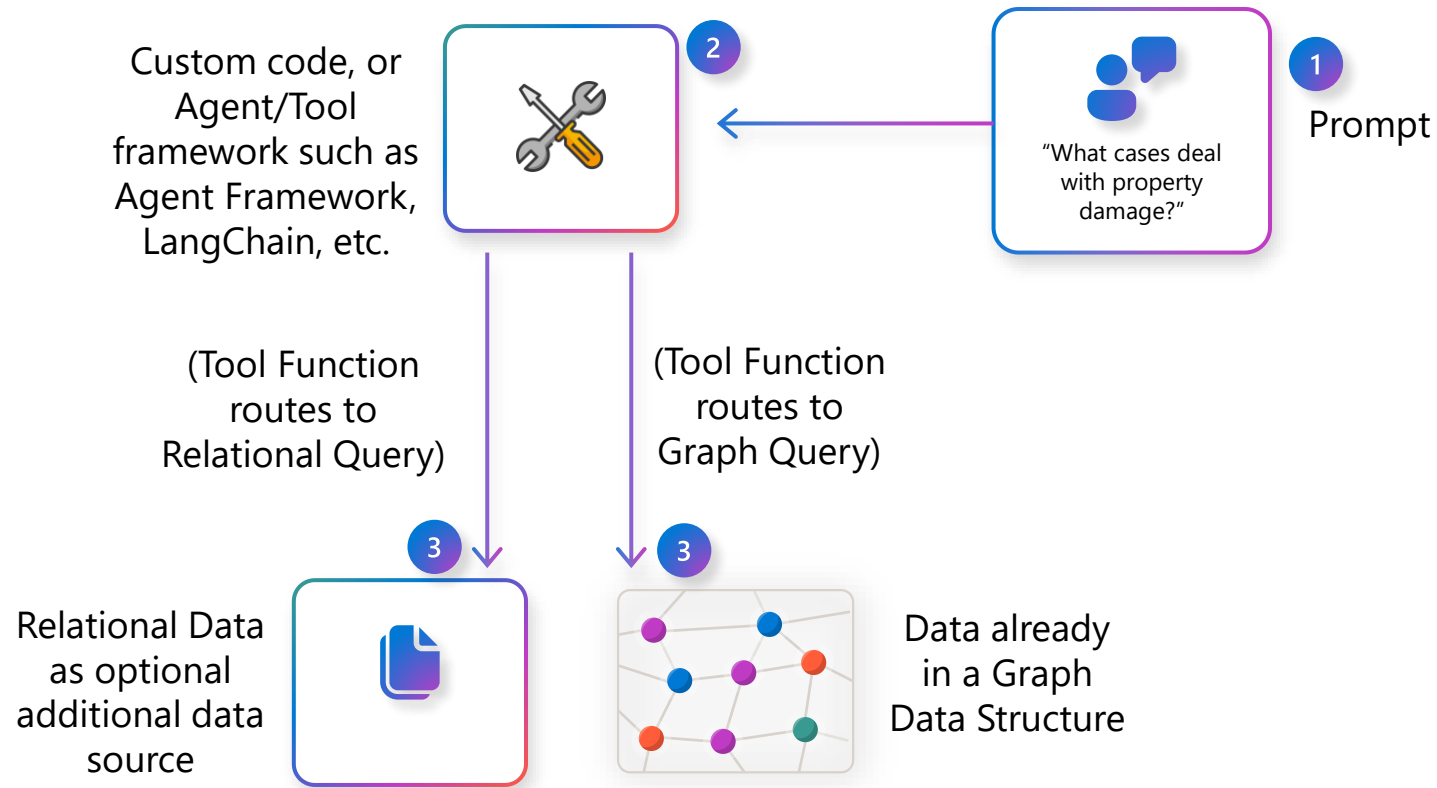


Parallel Tool Call

- Faster
- Used for independent tasks

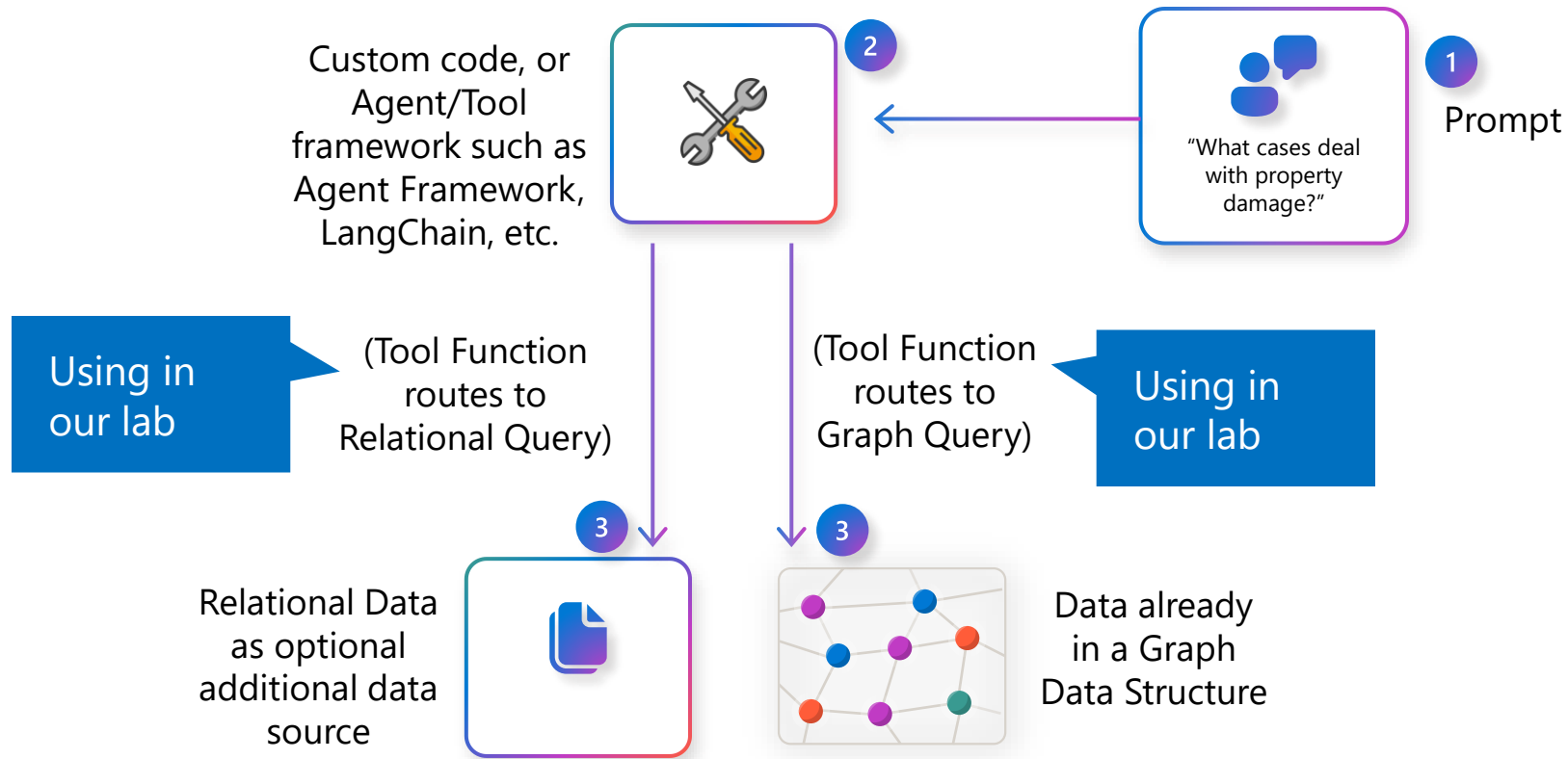
GraphRAG – Example

GraphRAG via Native Graph Data Querying



GraphRAG – Example

GraphRAG via Native Graph Data Querying



AI Functions

<code>azure_ai.generate()</code>	Generates text or structured output using Large Language Models (LLMs).
<code>azure_ai.is_true()</code>	Evaluates the likelihood that a given statement is true.
<code>azure_ai.extract()</code>	Extracts structured features or entities from text.
<code>azure_ai.rank()</code>	Reranks a list of documents based on relevance to a given query.

AI Functions

`azure_ai.generate()` Generates text or structured output using Large Language Models (LLMs).

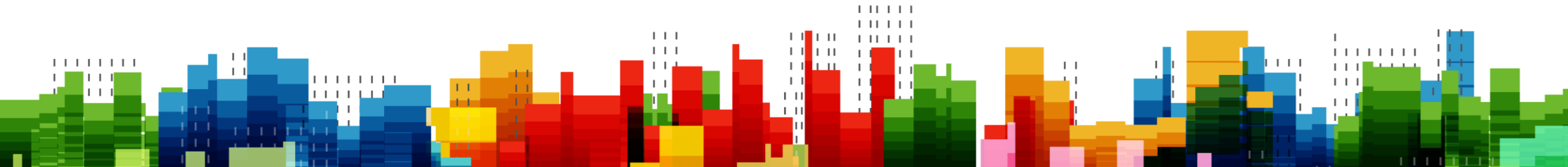
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`azure_ai.extract()` Extracts structured features or entities from text.

`azure_ai.rank()` Reranks a list of documents based on relevance to a given query.

Using in
our lab

Lab “Follow Me”



Related Sessions

BRK223

From rows to reasoning: Designing databases for AI apps and agents

Date: Tuesday, June 2

Time: 2:30 PM - 3:15 PM

Location: Festival Pavilion, Breakout 2

LTG413

Cloud-Native PostgreSQL, Rebuilt for Scale: Azure HorizonDB

Date: Tuesday, June 2

Time: 4:50 PM - 5:05 PM

Location: Gateway Pavilion, Level 2, Theater 1

TT662

Scaling PostgreSQL for Agentic Apps: Patterns and Tradeoffs

Date: Tuesday, June 2

Time: 5:15 PM - 6:00 PM

Location: Gateway Pavilion, Level 2, Table 2

DEM364

Simplify app dev with cloud-native PostgreSQL in Azure HorizonDB

Date: Wednesday, June 3

Time: 2:30 PM - 3:15 PM

Location: Gateway Pavilion, Level 2, Theater B



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Lab GitHub Repo

(Long term)



github.com/jjfrost/build-2026-lab

Thank you!

